Sharif University of Technology: EE Seminars

Information-Theoretic Analysis of Learning Algorithms

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Overview of Talk

Goal: Analysis of the performance of learning algorithms using tools from information theory and communication theory.

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1. Quick Review of Information Theory
2. Bounding Generalization Gap
3. Gibbs' Algorithm and Noisy ERM
4. Tight Bounds: Chaining
5. Connections to VC Theory
6. Minimum Excess Risk

Quick Review of Information Theory

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Let Q be a probability distribution on the discrete set $\mathcal{X}$. Entropy is defined as

$$H(Q) = -\sum_{x \in \mathcal{X}} q(x) \log(q(x)). \tag{1}$$

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Differential Entropy

Let Q be a probability distribution on the set $\mathcal{X}$ with density q wrt. Lebesgue measure. Entropy is defined as

$$H(Q) = -\mathbb{E}\big[\log q(X)\big] = -\int_{\mathcal{X}} q(x) \log(q(x)) dx. \tag{2}$$

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Shannon's First Theorem (Shannon 1948)

(Informal Statement)

Let $X_1, \ldots, X_n \sim P_X$ be n i.i.d. variables with entropy $H(P_X)$.

- This sequence can be compressed into $nH(X)$ bits as $n \to \infty$.
- If compressed to less than $nH(X)$ bits, information is lost.

Problem Formulation

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  Hypothesis Set: $\mathcal{W}$
  Loss Function: $\ell : \mathcal{W} \times \mathcal{Z} \to \mathbb{R}^+$

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- Given $S = (Z_1, \ldots, Z_n) \sim \mu$.
- For every $w \in \mathcal{W}$, define

$$\begin{cases} L_\mu(w) = \mathbb{E}_\mu[\ell(w, Z)] = \int \ell(w, z)\mu(dz) \\ \\ L_S(w) = \frac{1}{n}\sum_{i=1}^{n} \ell(w, Z_i) \end{cases}$$

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- **Goal**: Algorithm picks $W \in \mathcal{W}$ according to some $P_{W|S}$. Control

$$\mathbb{E}[\mathrm{gen}(W)] = \mathbb{E}[L_\mu(W) - L_S(W)],$$

where the expectation is over $P_{S,W} = \mu^{\otimes n} P_{W|S}$.

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- These bounds depend only on the hypothesis set: **pessimistic**.
- **Intuition**:

$$\text{less information usage from } S \implies \text{less overfitting}$$

Decoupling Estimate

Definition

The random variable X is called σ-subgaussian if

$$\forall \lambda \in \mathbb{R}: \quad \mathbb{E}\left[e^{\lambda(X-\mathbb{E}[X])}\right] \leq e^{\lambda^2\sigma^2/2}.$$

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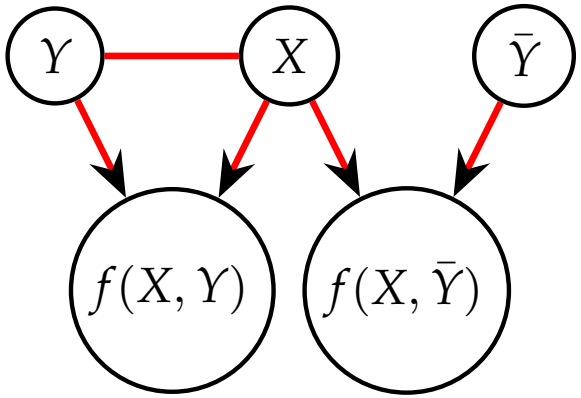
Lemma [Xu and Raginsky, 2017] & [Russo and Zou, 2016]

Let $(X, Y) \sim P_{XY}$, and $\bar{Y} \sim P_Y$ be independent copy. For any real valued $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, if $f(X, \bar{Y})$ is σ-subgaussian, then

$$\left|\mathbb{E}[f(X, Y)] - \mathbb{E}[f(X, \bar{Y})]\right| \leq \sqrt{2\sigma^2 I(X; Y)}$$



Decoupling Estimate

Proof: Donsker-Varadhan variational representation:

$$\mathrm{KL}(\pi||\rho) = \sup_F \Big\{ \int_\Omega F d\pi - \log \int_\Omega e^F d\rho \Big\}.$$

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For any $\lambda \in \mathbb{R}$, we have

$$\begin{aligned}\mathrm{KL}(P_{XY}||P_X \otimes P_Y) &\geq \mathbb{E}[\lambda f(X,Y)] - \log \mathbb{E}[e^{\lambda f(X,\bar{Y})}] \\ &\geq \lambda \mathbb{E}[f(X,Y)] - \lambda \mathbb{E}[f(X,\bar{Y})] - \frac{\lambda^2\sigma^2}{2}\end{aligned}$$

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Discriminant must be non-positive, which concludes the proof. $\square$

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Theorem [Xu and Raginsky, 2017]

Suppose that $\ell(w, Z)$ is σ-subgaussian for μ, under all $w \in \mathcal{W}$. We have

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Remark

- The learning algorithm $P_{W|S}$: **channel** from S to W.
- $\sup_\mu I(S; W)$: channel **capacity** of the channel, under the constraint that the input distribution is of a product form.

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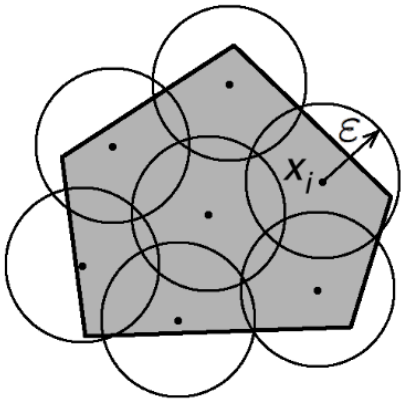

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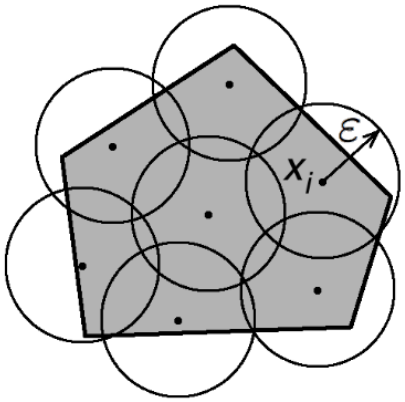

Dudley bound: multi-scale approximation of T

$$\mathbb{E}\left[\sup_{t\in T} X_t\right] \leq 6\sum_{k\in\mathbb{Z}} 2^{-k}\sqrt{\log \mathcal{N}(T,d,2^{-k})}.$$

Chaining Mutual Information

Chaining Mutual Information [Asadi, Abbe & Verdu. 2019]

If $\{\mathrm{gen}(w)\}_{w\in\mathcal{W}}$ is a subgaussian process in $(\mathcal{W}, d)$:

$$\mathbb{E}[\mathrm{gen}(W)] \leq 3\sqrt{2} \sum_{k=k_1(\mathcal{W})}^{\infty} 2^{-k}\sqrt{I(\pi_k(W); S)}.$$

Bayesian Setting

Bayesian Learning

- **Generation Model**:

$$P_{W,S,Z} = P_W \otimes \prod_{i=1}^{n} P_{Z_i|W} \otimes P_{Z|W}$$

$$\forall\, i \in [n], \quad P_{Z_i|W} = P_{Z|W}$$

- *Predicting Modeling Framework*: $Z = (X, Y)$, and $Z_i = (X_i, Y_i)$.

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- *Predicting Modeling Framework*: $Z = (X, Y)$, and $Z_i = (X_i, Y_i)$.

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- **Goal**: predict Y based on X and observations $S = \{Z_1, \ldots, Z_n\}$.

Minimum Excess Risk

- **Generalized Entropy**:

$$R_\ell(Y|X) = \inf_{\psi:\mathcal{X}\to\mathcal{Y}} \mathbb{E}[\ell(Y, \psi(X)]$$

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$$\mathrm{MER}^n_\ell = R_\ell(Y|S,X) - R_\ell(Y|W,X)$$

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- MER is algorithm *independent*.

Theorem [Xu & Raginsky 2020], [Hafez & Moniri, 2021]

The following bound can be derived for MER:

$$\mathrm{MER}^n_\ell \leq \sqrt{\frac{b^2}{2} I(Y;W|S,X)}.$$

Minimum Excess Risk: Lower Bounds

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Remark [Hafez & Moniri 2020]

It is *impossible* to find a matching lower bound such that

$$\mathrm{MER}_{\ell}^{n} \geq \alpha \sqrt{I(Y; W|S, X)}.$$

Minimum Excess Risk: Lower Bounds

- Define the following distortion function:

$$d(w, \hat{h}(.)) = \mathbb{E}_{XY|W=w}[\ell(Y, \hat{h}(X)) - \ell(Y, \psi^*_{Y|XW}(w, X))].$$

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- **Optimal algorithm**: outputs $\hat{h}(.) = \psi^*_{Y|SX}(s, .)$.

$$\begin{aligned}
&\mathbb{E}_{WS}[d(W, \psi^*_{Y|SX}(S, .))] \\
&\quad = \mathbb{E}_{WSXY}[\ell(Y, \psi^*_{Y|SX}(S, X)) - \ell(Y, \psi^*_{Y|WX}(W, X))] \\
&\quad = R_\ell(Y|S, X) - R_\ell(Y|W, X) = \mathrm{MER}^n_\ell.
\end{aligned}$$

Minimum Excess Risk: Lower Bounds

- Define the (constrained) rate-distortion optimization:

$$D_n(R) = \inf_{P_{\hat{h}|S}} \mathbb{E}[d(W, \hat{h})], \quad \text{s.t.} \;\; I(W; \hat{h}) \leq R.$$

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- Note that $W \to S \to \hat{h}$ and $P_{S|W}$ is fixed.

Theorem

For a given training set size n, for all rates $R \geq I(W; S)$, we have

$$D_n(R) = \mathrm{MER}_\ell^n.$$

Explicit Bounds on MER

Assume that $\mathcal{W}$ is a d-dimensional subset of $\mathbb{R}^d$.

- **Upper Bounds** under some regularity conditions on $P_{Z|W}$:
  - $\mathrm{MER}_l^n = O(\frac{1}{n})$ for bounded loss.
  - $\mathrm{MER}_2^n = O(\frac{1}{n})$ for quadratic loss.

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- **Lower Bounds** using the R/D view and the Shannon Lower Bound, in some cases, we prove $\Omega(\frac{1}{n})$ rates.
- For example, in $Y = W^\top X + \sigma\nu$ with

$$\begin{cases} W \sim \mathcal{N}(0, I_{p\times p}) \\ X \sim \mathcal{N}(0, \Sigma_X) \\ \nu \sim \mathcal{N}(0, I_{p\times p}) \end{cases}$$

  we have $\mathrm{MER}_2^n = \Omega(\frac{p}{n})$.

Conclusion

Using information theoretic tools:

- We upper bounded generalization gap in a frequentist setting.
- We upper and lower bounded minimum excess risk in Bayesian statistics.

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